

Thermin and Norton Equivalents

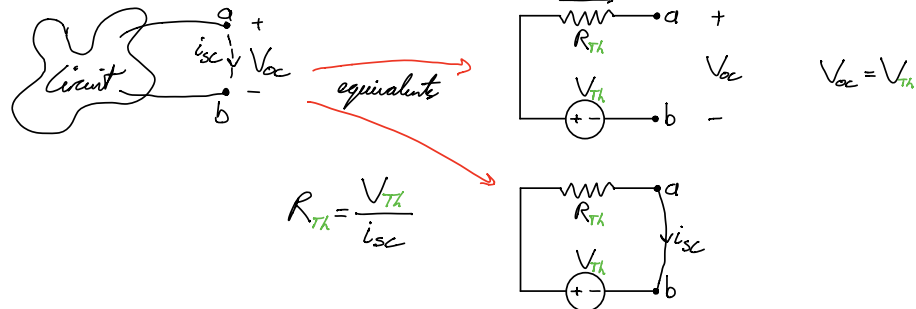
- replace entire circuit, other than two nodes, with a **resistor** and **voltage** source in **series**
- replace entire circuit, other than two nodes, with a **resistor** and **current** source in **parallel**

Elements

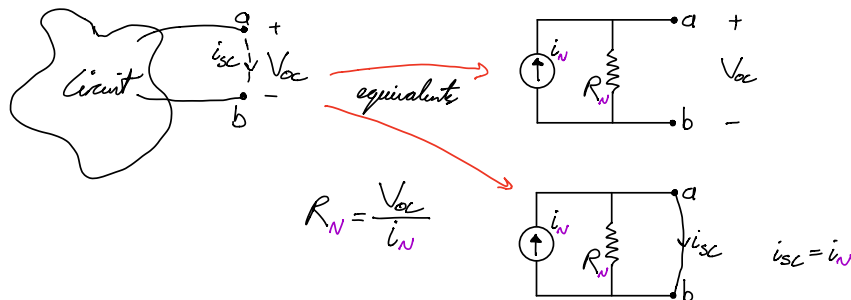
- Open Circuit: $i_{oc} = 0 \rightarrow V_{oc}$
- Short Circuit: $V_{sc} = 0 \rightarrow i_{sc}$

- Ideal Ammeter: short circuit that displays **current**
- Ideal Voltmeter: open circuit that displays **voltage**

Thermin (V_{Th}, R_{Th})



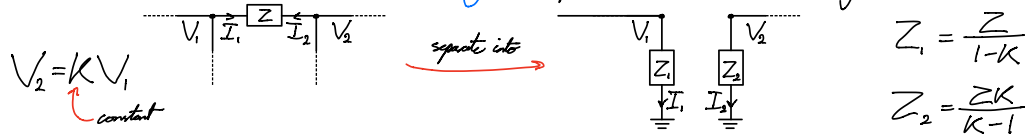
Norton (I_N, R_N)



- if all sources are **independent**, kill them and calculate: $R_{eq} = R_{Th}$
- with **dependent** sources, calculate open circuit voltage (V_{oc}), and short circuit current (i_{sc})

Miller's Theorem

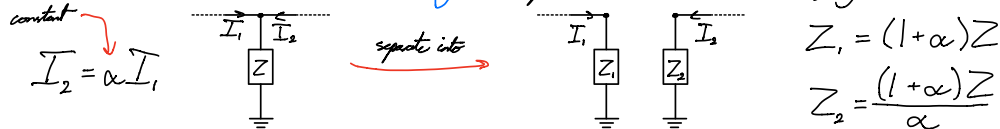
- used to separate a circuit at a certain impedance element
- given a network with a branch, containing an impedance (Z), connecting 2 nodes (V_1, V_2):



- Z_1 draws same current (I_1) from V_1 as Z did, similar phenomenon with Z_2

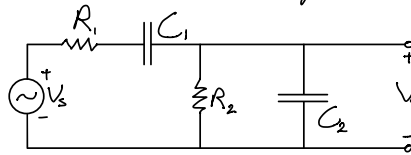
$$I_1 = \frac{V_1 - V_2}{Z} = V_1 \frac{1-K}{Z} = -I_2$$

- given a network with a branch, connecting an impedance (Z) to 2 converging currents (I_1, I_2):

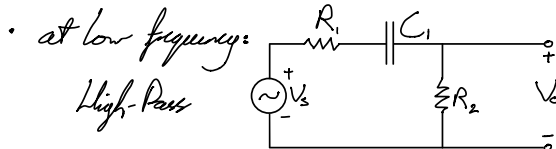


Band Pass / High Pass / Low Pass

- this circuit will behave differently depending on the input frequency



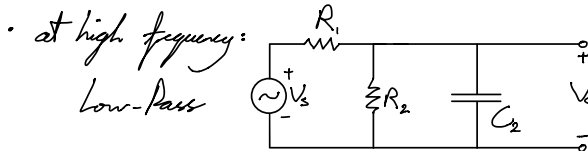
- depending on C_1 & C_2 , this could be a band pass or low pass filter
- $C_1 \gg C_2$ for band pass



- at low frequency:

High-Pass

$$\omega_{LP} = \frac{1}{RC} \leftarrow \text{low frequency pole}$$



- at high frequency:

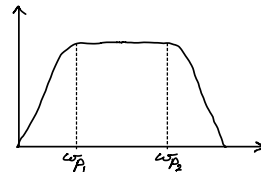
Low-Pass

$$\omega_{HP} = \frac{1}{RC} \leftarrow \text{high frequency pole}$$

- the Transfer Function consists of 2 poles and a zero at $s=0$:

$$H(s) = \frac{\alpha s}{(s + \omega_{P1})(s + \omega_{P2})}$$

- the poles border the midband region:



- the transfer function can be separated into 3 regions: midband highpass lowpass

$$H(s) = \frac{R_2}{R_1 + R_2} \cdot \frac{s}{(s + \omega_{P1})} \cdot \frac{\omega_{P2}}{(s + \omega_{P2})} = A_m \cdot F_L(s) \cdot F_H(s)$$

$$F_L(s) = \frac{(s + \omega_{Lz1}) \cdots (s + \omega_{Lzn})}{(s + \omega_{Lp1}) \cdots (s + \omega_{LpN})}$$

$$N = n$$

$$F_H(s) = \frac{\omega_{Hp1} \cdots \omega_{HpM}}{(s + \omega_{Hp1}) \cdots (s + \omega_{HpM})} \cdot \frac{(s + \omega_{Hz1}) \cdots (s + \omega_{HzM})}{\omega_{Hz1} \cdots \omega_{HzM}}$$

$$M > m$$

— Zeros and Poles

- use **zero and pole locations** to find cut-off frequencies

$$\omega_{LC} \doteq \sqrt{\omega_{Lp1}^2 + \cdots + \omega_{LpN}^2 - 2\omega_{Lz1}^2 - \cdots - 2\omega_{Lzn}^2}$$

$$\omega_{HC} \doteq \left[\omega_{Hp1}^{-2} + \cdots + \omega_{HpM}^{-2} - 2\omega_{Hz1}^{-2} - \cdots - 2\omega_{Hzm}^{-2} \right]^{-\frac{1}{2}}$$

- using **Open Circuit & Short Circuit Time Constant Approximation** to find cut-off frequencies

- can be used for single **high pass** and/or **low pass** poles
- poles must be at least a **factor of 4** away from each other

- Low Frequency Poles ($\omega_{LP} = 1/\tau_{LP}$)

- this only applies to capacitors of the **high pass sub-circuit** → **open circuit** the **low pass** capacitors

1. find the pole with the **highest frequency**: ω_{Lpa}
 - calculate poles by **short circuiting** other poles

2. **open circuit** the capacitor of this pole ($C_a \rightarrow 0$) and recalculate the poles

3. find the pole with the **next highest frequency**: ω_{Lpb}

4. repeat 2 and 3 until lowest frequency pole is reached: ω_{Lpx}

5. determine ω_{LC} by combining all poles in equation prescribed above

- High Frequency Poles ($\omega_{HP} = 1/\tau_{HP}$)

- this only applies to capacitors of the **low pass sub-circuit** → **short circuit** the **high pass** capacitors

1. find the pole with the **lowest frequency**: ω_{Hpa}
 - calculate poles by **open circuiting** other poles

2. **short circuit** the capacitor of this pole ($C_a \rightarrow \infty$) and recalculate the poles

3. find the pole with the **next lowest frequency**: ω_{Hpb}

4. repeat 2 and 3 until lowest frequency pole is reached: ω_{Hpx}

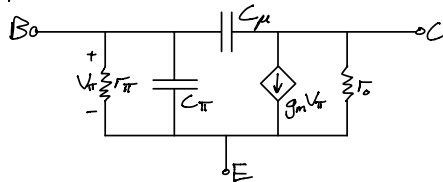
5. determine ω_{LC} by combining all poles in equation prescribed above

BJT as an Amplifier

- **small signal** operation describes an AC signal superimposed on a DC steady state signal
- to function as amplifier, transistor must be biased in its **active region**; such that:

$$V_{BE} \approx 0.7V \quad V_{CE} > 0V$$

- Hybrid- π Model:



$$g_m = \frac{I_C}{V_T}$$

$$r_{\pi} = \frac{V_T}{I_B}$$

$$r_o = \frac{V_A}{I_C}$$

- Transistor DC Relationships:

$$I_C = \alpha I_E = \beta I_B = I_S e^{\frac{V_{BE}}{V_T}}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

$$r_{\pi} = \frac{V_{be}}{i_b} = \frac{\beta}{g_m}$$

$$I_E = I_C + I_B = (1 + \beta) I_B$$

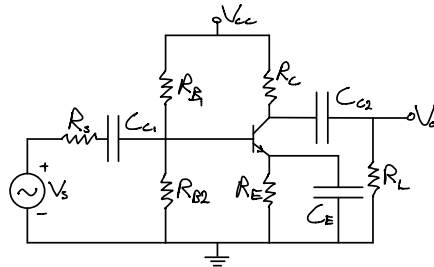
$$\beta = \frac{\alpha}{1 - \alpha}$$

$$i_c = i_b \cdot \beta = g_m V_{be}$$

small-signal transconductance $\uparrow$

- Common Emitter Amplifier

- possesses low input and high output impedance
- generates large and inverting power gain to maximize power transfer
- bias circuit with **Coupling** (C_{C1}, C_{C2}) and **Bypass** (C_E) capacitors
 - V_S/R_S is **Source**, R_L is **Load**

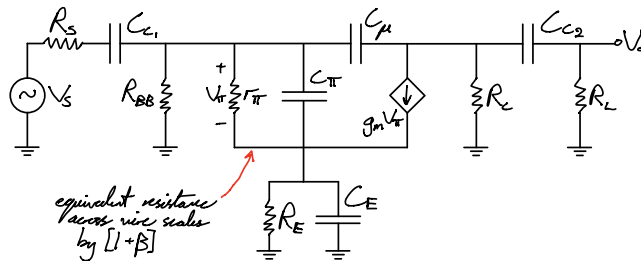


- bias the transistor using **one** of the **1/3 rules**:

$$1. \quad V_B = \frac{V_{CC}}{3} \quad V_C = \frac{2V_{CC}}{3} \quad I_1 = \frac{I_E}{\beta}$$

$$2. \quad V_E = \frac{V_{CC}}{3} \quad V_C = \frac{2V_{CC}}{3} \quad I_1 = \frac{I_E}{\beta}$$

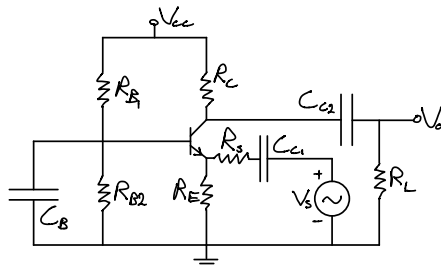
- Complete Small Signal Model



$$\text{zero at: } \omega_{zCE} = \frac{1}{R_E C_E}$$

Common Base Amplifier

- possesses very low input impedance to maximize power transfer
- bias circuit with **Coupling** (C_{c1}, C_{c2}) and **Bypass** (C_B) capacitors
- V_S/R_S is **Source**, R_L is **Load**

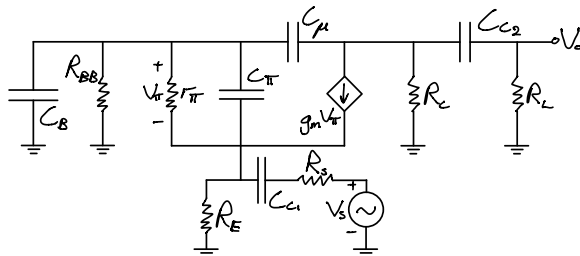


• bias the transistor using **one** of the **V_B rules**:

$$1. \quad V_B = \frac{V_{CC}}{3} \quad V_C = \frac{2V_{CC}}{3} \quad I_1 = \frac{I_E}{\beta}$$

$$2. \quad V_E = \frac{V_{CC}}{3} \quad V_C = \frac{2V_{CC}}{3} \quad I_1 = \frac{I_E}{\beta}$$

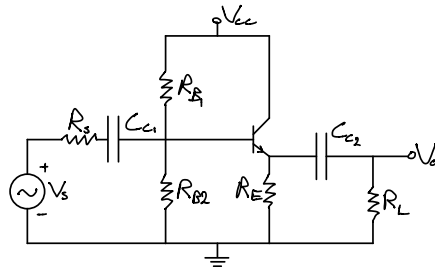
Complete Small Signal Model



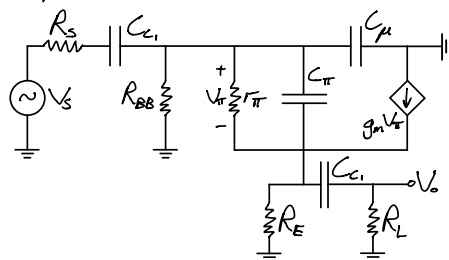
$$\text{zero at: } \omega_{L_{C_B}} = \frac{1}{R_{BB}C_B}$$

Common Collector Amplifier

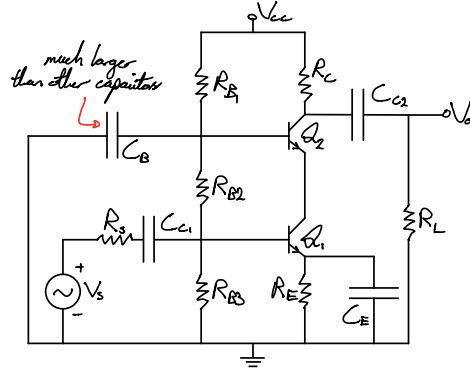
- creates high current gain and wide bandwidth
- usually used as **output** of multi-stage amplifier
- bias circuit with **Coupling** (C_{c1}, C_{c2}) capacitors



Complete Small Signal Model



- Cascode Amplifier
 - provides higher bandwidth and reduced reverse transmission
 - bias circuit with **Coupling** (C_{C1}, C_{C2}) and **Bypass** (C_E, C_B) capacitors

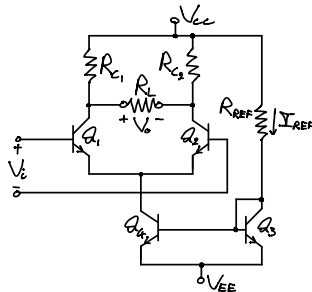


- bias the transistors using the $\frac{1}{4}$ rule:

$$V_{C2} = \frac{3V_{CC}}{4} \quad V_{E1} = \frac{V_{CC}}{4}$$

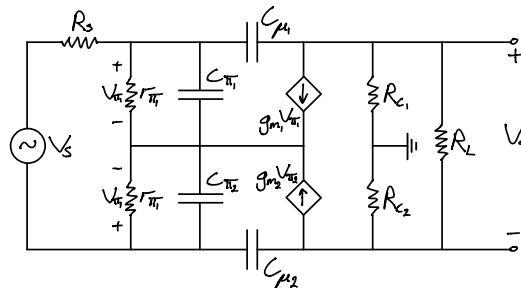
$$V_{E2} = V_{C1} = \frac{V_{CC}}{2} \quad I_1 = \frac{I_E}{\beta}$$

- Differential Amplifier
 - this circuit does not require Bypass capacitors to establish DC bias



$$A_m = -g_m R_c \left[\frac{2r_{\pi}}{2r_{\pi} + R_S} \right] \left[\frac{R_L}{R_L + 2R_c} \right]$$

- Complete Small Signal Model

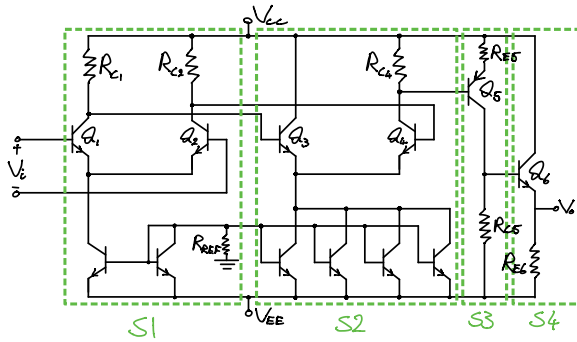


$$\omega_{Hpl} = \left[\left[\frac{C_{\pi}}{2} + \frac{C_{\mu}}{2} (1-k) \right] [2r_{\pi} // R_S] \right]^{-1}$$

$$\omega_{Hpl} = \left[\frac{C_{\mu}}{2} \left(1 - \frac{1}{k} \right) \cdot R_L // 2R_c \right]^{-1}$$

Operational Amplifiers

- this amplifier produces high gain using a *differentiated input*
- Basic Op Amp Circuit



- 4 stages of this Amplifier:

S1: Input Stage

S2: Gain Stage

S3: Level Shifter

S4: Output Stage

$$A_{M1} = -\frac{g_{m1}}{2} \cdot (R_{C1} + R_{C2}) \cdot \frac{r_{\pi3} + r_{\pi4}}{r_{\pi3} + r_{\pi4} + R_{C1} + R_{C2}}$$

$$A_{M2} = \frac{g_{m3}}{2} \cdot R_{C4} \parallel (r_{\pi6} + \beta R_{E5})$$

$$A_{M3} = -\frac{R_{C5} \parallel \beta R_{E6}}{R_{E5}}$$

$$A_{M4} = 1$$

- Skew Rate

- maximum rate of change of output voltage: $\frac{dV_o}{dt} \max = \omega_E V_i$
max frequency before distortion

- Full Power Bandwidth

- maximum power delivered to load before distortion
- occurs at maximum output voltage: $\omega_m V_{max} = \omega_E V_i$

- Common Mode Rejection Ratio

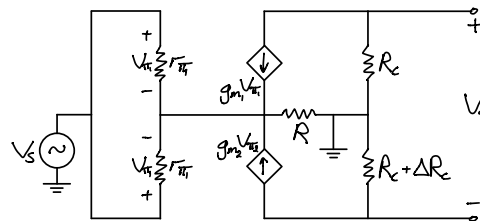
- represents ability of amplifier to reject common-mode signals
signal present in both input terminals

$$CMRR = \frac{|A_d|}{|A_{cm}|}$$

← differential gain *← common mode gain*

- single op amp small signal model:

$$CMRR = g_m \cdot 2R \cdot \frac{R_c}{\Delta R_c}$$



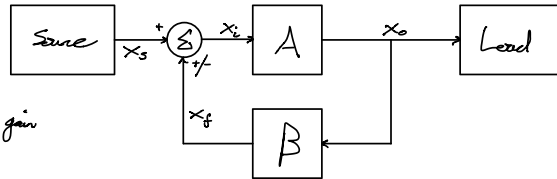
Feedback

- output is linear combination of input and *previous output*
- reduces noise variance $\rightarrow$ higher signal stability

- gain of feedback amplifier:

$$A_f = \frac{A}{1 + A\beta}$$

amount of feedback $\rightarrow$ loop gain



- Desensitivity Factor $(1 + A\beta)$

- factor by which the voltage gain is reduced due to feedback network

- Bandwidth Extension: decrease midband gain and increase bandwidth by factor of: $[1 + A_m\beta]$

$$A = A_m \cdot \frac{\omega_H}{s + \omega_H} \rightarrow A_f = A_m \frac{\omega_H}{s + \omega_H(1 + A_m\beta)}$$

- Impedance Control

- input and output impedance is affected by amount of feedback

- Series-Shunt: $R_{if} = R_i(1 + A\beta)$ $R_{of} = \frac{R_o}{1 + A\beta}$
- Shunt-Series: $R_{if} = \frac{R_i}{1 + A\beta}$ $R_{of} = R_o(1 + A\beta)$
- Shunt-Shunt: $R_{if} = \frac{R_i}{1 + A\beta}$ $R_{of} = \frac{R_o}{1 + A\beta}$
- Series-Series: $R_{if} = R_i(1 + A\beta)$ $R_{of} = R_o(1 + A\beta)$

- Feedback Network (β -network) Parameters

- Series-Shunt: $\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$ $h_{11} = \frac{V_1}{I_1} \big|_{V_2=0}$ $h_{21} = \frac{I_2}{I_1} \big|_{V_2=0}$
Voltage Amplifier

$$h_{12} = \frac{V_1}{V_2} \big|_{I_1=0} \quad h_{22} = \frac{I_2}{V_2} \big|_{I_1=0}$$

- Shunt-Series: $\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ $g_{11} = \frac{I_1}{V_1} \big|_{I_2=0}$ $g_{21} = \frac{V_2}{V_1} \big|_{I_2=0}$
Current Amplifier

$$g_{12} = \frac{I_1}{I_2} \big|_{V_1=0} \quad g_{22} = \frac{V_2}{I_2} \big|_{V_1=0}$$

- Shunt-Shunt: $\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$ $y_{11} = \frac{I_1}{V_1} \big|_{V_2=0}$ $y_{21} = \frac{I_2}{V_1} \big|_{V_2=0}$
Transmittance Amplifier

$$y_{12} = \frac{I_1}{V_2} \big|_{V_1=0} \quad y_{22} = \frac{I_2}{V_2} \big|_{V_1=0}$$

4. Series-Series: $\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$
 $z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}$
 $z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}$
 Transconductance Amplifier
 $z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}$
 $z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}$

- Stability
 - frequency range of amplifier to prevent exponential growth: $|A(s)B(s)| \in (0, 1]$